

Math 541
Homework 4

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Exercise 1. The Klein bottle K is the space obtained from a square by means of the labelling scheme $aba^{-1}b$. Describe a two sheeting covering map $p : \mathbb{T}^2 \rightarrow K$, where $\mathbb{T}^2$ is the torus.

Proof.

□

Exercise 2. What compact surfaces cover the Klein bottle? Justify your answer.

Proof. Suppose $p : E \rightarrow K$ is an n -sheeted covering space of the Klein bottle. Then $\chi(E) = n \cdot \chi(K) = n \cdot 0 = 0$. It must be the case that E is either a Torus or a Klein bottle. □

Exercise 3. Let N_g be a compact non-orientable surface of genus $g > 1$. We are given that there exists a two sheeted covering map $p : S \rightarrow N_g$, where S is a compact orientable surface. What is the genus of S ? Justify your answer.

Proof.

□

Exercise 4. Let S_3 denote the compact orientable surface of genus 3. Assume that, for every positive integer n there exists an n -sheeted connected covering space of S^3 . State which compact surfaces cover S_3 . Justify your answer.

Proof.

□

Exercise 5. Prove that a non-trivial finite group is not a free group.

Proof. We will prove the converse: let F be a free group on a set X . We will show that every nonidentity element in F has infinite order, establishing that F must be an infinite group. Suppose $x_1^{\delta_1} \dots x_n^{\delta_n}$ is a nonidentity element with order $k > 1$. Then:

$$\underbrace{(x_1^{\delta_1} \dots x_n^{\delta_n}) \dots (x_1^{\delta_1} \dots x_n^{\delta_n})}_{k\text{-times}} = 1.$$

For this to be true, it must be the case that $x_n^{\delta_n} x_1^{\delta_1} = 1$. This implies that the word $x_2^{\delta_2} \dots x_{n-1}^{\delta_{n-1}}$ also has order $k > 1$. We continue this process inductively, and ultimately show that $x_i^{\delta_i} x_{n-i+1}^{\delta_{n-i+1}} = 1$. Since inverses commute, we can reduce our original word down to the

identity 1 from the inside out. This contradicts our claim that $x_1^{\delta_1} \dots x_n^{\delta_n}$ is a nonidentity element.

□

Exercise 6. Use the universal property of free products to prove that $\mathbb{Z} * (\mathbb{Z}/n\mathbb{Z})$ is non-abelian, for every integer n greater than 1.

Proof.

□

Exercise 7.

- (a) Let H and K be conjugate subgroups of a group G . Prove that if f is any homomorphism of G into some other group such that $f(H) = 1$, then $f(K) = 1$ also.
- (b) Let $G = A * B$, where A and B are nontrivial groups. Prove that A and B are not conjugate. [Hint: Use part (a) and the universal property of free products.]

Proof.

□